MIL-D-46209C (PA)
21 April 1975
Superseding
MIL-D-46209B (MU)
29 February 1968 and
MIL-D-46210A (MU)
25 January 1965
(See 6.6)

MILITARY SPECIFICATION

DETONATOR, STAB, M44E1, PARTS FOR, AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for and the loading, assembling and packing for one type of detonator designated as Detonator, Stab, M44E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Packaging, Methods of.
*MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

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MIL-STD-650 - Explosive: Sampling, Inspection and Testing.
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

*(For meaning of asterisk (*) see paragraph 6.5).

DRAWINGS

U.S. ARMY PICATINNY ARSENAL

*PRODUCT AND PACKING DRAWINGS

7548254 - Detonator, M44E1, Loading Assembly.
9297861 - Disc, Detonator Closing.
9297860 - Cup, Detonator.
*9258182 - Box, Set-Up, Packing Ammunition, Inner For Small Explosive Components.
*9258183 - Box, Set-Up, Packing, Ammunition, Outer for Small Explosive Components.
*9258184 - Box, Packing, Ammunition For Small Explosive Components.

*INSPECTION EQUIPMENT DRAWINGS

9200390 - Firing Assembly.
8816367 - Guide, Firing Pin.
9200012 - Pin, Firing.
8816015 - Ball, Steel.
8827828 - Sleeve.
8816368 - Disc, Lead.
8827829 - Bushing.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

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ASTM Designation E300 - Industrial Chemicals, Recommended Practice for Sampling

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Penn. 19103).

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts. - The parts shall comply with all requirements specified on Drawings (dwgs.) 9297860 and 9297861 as applicable and with all requirements specified in applicable specifications.

3.3 Detonator. - The detonator shall comply with all requirements specified on Drawing (dwg) 7548254 and with all requirements specified in applicable specifications.

3.4 Moisture content.

3.4.1 Lead azide. - The moisture content of the lead azide used in loading the detonator shall not exceed 0.50 percent.

3.4.2 Primer mixture. - The moisture content of the primer mixture used in loading the detonator shall not exceed 0.30 percent.

3.5 Functioning.

3.5.1 Output. - The detonator shall function on the first attempt when an energy of 119.25 inch ounces, maximum (max) is applied, and shall cause perforation of a 0.1345 inch thick lead disc, to a diameter of not less than 0.125 inch.

3.5.2 Waterproofness. - The detonator shall comply with the requirements of 3.5.1 after being submerged in water to a minimum depth of 2 inches for not less than 48 hours, at a temperature of 70 degrees Fahrenheit (F) plus or minus 10 degrees.

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*3.6 Workmanship. - All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

*3.7 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

*4.2 Classification of inspections. - The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

*4.3.1 Submission. - The contractor shall submit a first article sample (see 6.1b) as designated by the Contracting Officer. The first article sample shall consist of fifty (50) complete sets of parts (i.e. fifty (50) of each and every component). The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3.

*4.3.2 Inspections to be performed. - See MIL-A-48078, paragraphs 4.4.2.1, 4.4.2.2 and Table I specified herein.

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TABLE I

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-46209C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*	Detonator, Stab, M44E1, Metal Parts				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Metal Parts (Dwgs. 9297860 and 9297861) Examination for defects	50(a)		3.2	4.4.2.1 and 4.4.2.2

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4.3.3 Rejection. - (See MIL-A-48078).

4.4 Quality conformance inspection.

*4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of detonators shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Lead azide of the same type from not more than two lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

c. Tetryl from not more than one lot.

d. Primer mixture from one batch or blending of batches which results in homogeneity of all primer mixture used in loading the entire detonator lot.

4.4.2 Examination. - (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTIONS

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46209C(PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.1	Disc Detonator Closing	1 OF 1	9297861
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major B					
131	Diameter		0.40%	3.2	Gage
132	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46209C(PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.2	Cup, Detonator	1 OF 1	9297860
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Crack or split		100.0%	3.2	Visual
Major B					
131	Inside diameter, max		0.40%	3.2	Gage
132	Outside diameter, min		0.40%	3.2	Gage
133	Thickness through bottom, min		0.40%	3.2	Gage
134	Length		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46209C(PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.3	Assembly	1 OF 1	7548254
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup cracked or split		100.0%	3.3	Visual
Major B					
131	Length		0.40%	3.3	Gage
132	Diameter		0.40%	3.3	Gage
133	Disc loose		0.40%	3.3	Manual
134	Lacquer seal missing or inadequate		0.40%	3.3	Visual
135	Color of lacquer incorrect		0.40%	3.3	Visual
136	Not crimped to full 360 degrees		0.40%	3.3	Visual
137	Disc punctured or torn		0.40%	3.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46209C(PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.4	Box, Set-Up, Inner, After Loading and Prior to Closing Slide Cover	1 OF 1	9258182
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Filler and cushion assembly missing		100.0%	3.3	Visual
2	Filler and cushion assembly improperly assembled		100.0%	3.3	Visual
3	Stiffner missing		100.0%	3.3	Visual
4	Slide cover assembled reversed		100.0%	3.3	Visual
Major A					
101	Release paper missing		0.40%	3.3	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.3	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46209C (PA)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
*4.4.2.5	Box, Set-Up, Outer, with Inner Set-Up Boxes Assembled and prior to Assembling Top	1	1	9258183
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Red surface on inner set-up boxes do not alternate on same side of outer set-up boxes		100.0%	3.3	Visual
2	Stiffner not facing one direction		100.0%	3.3	Visual
Major A					
101	Inner box does not fit properly in outer box		0.40%	3.3	Visual
102	Fillers missing		0.40%	3.3	Visual
Major B					
131	Label not securely applied to box		0.40%	3.3	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.2	Visual
202	Box improperly taped		0.65%	3.3	Visual
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46209C(PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.6	Sealed Packed Barrier Bag, Prior to Packing in Box	1 OF 1	9258183
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101	Waterproofness of barrier bag destroyed by rupture, puncture or separation of heat seal		0.40%	3.3	Visual
102	Evidence of crushed box		0.40%	3.3	Visual
NOTE:	Package should be opened if there is evidence of crushing				
Minor					
201	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied		0.65%	3.3	Visual
202	Marking of barrier bag misleading or unidentifiable		0.65%	3.3	Visual
203	Evidence of poor workmanship		0.65%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46209C (PA)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
*4.4.2.7	Box, Packing, Ammunition Prior to Assembling Cover and Strapping	1	1	9258184
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101	Bags containing set-up boxes damaged		0.40%	3.3	Visual
102	Cushioning material incomplete or missing		0.40%	3.3	Visual
103	Marking missing, misleading or unidentifiable		0.40%	3.3	Visual
104	Bottom cushioning missing or of incorrect thickness		0.40%	3.3	Visual
Minor					
201	Contents loose within box		0.65%	3.3	Visual-Manual
202	Evidence of poor workmanship		0.65%	3.3	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46209C(PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
*4.4.2.8	Box, Packing, Ammunition	1 OF 1	9258184
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101	Strapping missing, broken, loose or Improperly fastened		0.40%	3.3	Visual/Manual
102	Box damaged		0.40%	3.3	Visual
103	Marking missing, incomplete or illegible		0.40%	3.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.3	Visual

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4.4.3 Testing. (See Note)

NOTE: Chemical sampling. - A representative sample shall be randomly selected for the tests specified in 4.4.3. The procedure described in ASTM E300 should be used to withdraw the samples. The tests in 4.5 shall be performed using prescribed analytical procedure for replicate determinations given in standard analytical textbooks.

4.4.3.1 Moisture content (see Table II).

Table II

Material
Lead azide (see 3.4.1)
Primer mixture (see 3.4.2)

The contractor shall provide adequate controls to insure compliance with the requirements, and shall test for verification one sample of each material from each eight hours production of detonators. Composite samples shall not be used. If the moisture content of a sample exceeds the requirement, and loading has not begun, that quantity of material represented by the sample shall be rejected. If detonators have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all detonators loaded with the non-conforming material shall be rejected. The tests shall be performed as specified in 4.5.1.1 or 4.5.1.2 as applicable.

4.4.3.2 Functioning.

4.4.3.2.1 Output (see 3.5.1). - Major A defect. A sample of three hundred and fifteen (315) detonators shall be randomly selected from each lot. If three (3) or more detonators fail to comply with the specified requirements the lot shall be rejected. If one or two defectives occur, a second sample of three hundred and fifteen (315) detonators shall be selected and tested. If, in the combined first and second sample, a total of four (4) or more defectives occur, the lot shall be rejected. The test shall be performed as specified in 4.5.2.1.

4.4.3.2.2 Waterproofness (see 3.5.3). - Major A defect. A sample of twenty-five (25) detonators shall be selected at

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random from each lot. If one or more detonators fail to comply with the requirements the lot shall be rejected. The test shall be performed as specified in 4.5.2.2.

4.4.3.3 Graphite content of tetryl (see dwg 7548254). - Major A defect. The graphite content of each lot of tetryl shall be determined prior to loading. A sample of sufficient size to perform the test specified in 4.5.3 shall be selected. If the graphite content does not comply with the requirement, the lot, including any detonators loaded with the non-conforming tetryl, shall be rejected.

4.4.3.4 Composition of primer mixture (see dwg 7548254). - Major A defect. - The composition of the primer mixture of each batch or blend of batches shall be determined at the time of loading. A sample of sufficient size to perform the test specified in 4.5.4 shall be selected. If the batch or blend of batches of primer mixture fails to comply with the requirements, that quantity of primer mixture represented by the sample shall be rejected. Any detonators that have been loaded with the non-conforming mixture shall also be rejected.

4.4.3.5 Heat seal test of sealed bag (see dwg 9258183). - Major A defect. - Sampling, acceptance criteria and test procedures for this test shall be as specified in MIL-P-116.

4.4.4 Inspection equipment. - The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the 'Paragraph Reference/ Inspection method' column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs, in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and section 6.2 herein.

4.5 Test methods and procedures.

4.5.1 Moisture content.

4.5.1.1 Lead azide. - The moisture content of the lead azide shall be determined in accordance with Method 101.5 of MIL-STD-650 except that a two (2) gram (gm) sample shall be tested and the dish shall be covered with a sheet of polytetrafluorethylene when weighing.

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4.5.1.2 Primer mixture. - Transfer a known weight of between 0.4 to 0.7 gram of sample to a tared weighing dish approximately 55 millimeters (mm) in diameter and 30 mm in height. Place the dish and contents in an oven maintained at 55 to 65 degrees Centigrade (C) or in a desiccator containing anhydrous calcium chloride until constant weight is obtained. Calculate the loss in weight as the percentage of moisture.

4.5.2 Functioning.

4.5.2.1 Output. - The detonator shall be assembled to the applicable test fixture specified in 4.4.4, so that the firing pin will strike the sensitive end of the detonator. A steel ball 1 1/2 inch in diameter and weighing 7.95 ounces maximum, shall be dropped on the firing pin, to function the detonator, from a height of 15 inches, maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective. Method of inspection shall be in accordance with dwgs 9200390, 8816367, 9200012, 8816015, 8827828, 8816368 and 8827829.

4.5.2.2 Waterproofness. - The detonator shall be immersed under water as specified for the time specified. Within one hour after removal from the water, the detonator shall be subjected to the output test specified in 4.5.2.1. Any detonator which fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective.

4.5.3 Graphite content of tetryl. - An accurately weighed portion of 5 grams shall be transferred to a tared medium porosity sintered glass crucible. The sample shall be extracted with four 10 milliliter (ml) portions of acetone, maintaining each portion in contact with the residue for approximately 5 minutes before applying suction. Any lumps which may persist shall be broken up and the sides of the crucible washed down with each 10 ml portion of acetone to insure complete removal of the tetryl. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in a drying oven at 95 degrees Centrigrade (C) plus or minus 5 degrees C for 1/2 hour, cooled in a desiccator and weighed. The above extraction shall be repeated, drying and weighing with individual 10 ml portions of acetone until the loss in weight on subsequent extractions totals less than 0.005 gram. The weight of residue in the crucible shall be calculated as the percentage of total graphite.

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$$\text{Total percent of graphite} = \frac{100A}{B}$$

when:

A = weight of residue in crucible
B = weight of sample on a moisture-free basis

4.5.4 Composition of primer mixture. - The composition of the primer mixture shall be determined by performing the following tests in the sequence specified:

4.5.4.1 Barium nitrate. - An accurately weighed portion of approximately 0.5 gram of primer mixture shall be transferred to a tared medium-porosity, fritted glass crucible of 20 ml capacity. Three ml of lead azide saturated distilled water shall be added at a temperature of 50 degrees C plus or minus 2 degrees C, agitated for 1 minute, and, if necessary, lumps shall be broken up with a rubber policeman attached to a glass rod. Suction shall be applied. This procedure shall be repeated five times, making a total of six 3-ml extractions. The sample in the filtering crucible shall be rinsed three times with alcohol, dried in an oven maintained at 60 degrees C plus or minus 5 degrees C for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of barium nitrate in the sample on a dry basis as follows:

$$\text{Barium nitrate, percent} = \frac{(A-B)100}{W(1-.01M)}$$

where:

A = weight of crucible and sample.
B = weight of the crucible and sample after extractions with lead azide saturated water.
M = moisture content of sample obtained from 4.5.3 in percent.
W = weight of sample.

4.5.4.2 Lead styphnate.

4.5.4.2.1 Basic lead styphnate (when applicable). - The residue obtained in the determination of barium nitrate shall be extracted with six 5-ml portions of saturated ammonium acetate solution at about 25 degrees C, (hot saturated ammonium acetate shall not be used as it dissolves antimony sulfide to some extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately 2 minutes. The residue shall

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be washed with water until the filtrate coming through the crucible appears colorless. The filtrate and washings shall be quantitatively transferred to a 250 ml calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, a 2 ml aliquot of this solution shall be transferred to a 50 ml calibrated volumetric flask and diluted to the mark with distilled water. A portion of this solution shall be transferred to a corex glass spectrphotometric cell having a width of approximately 1 centimeter (cm) and its absorbance determined at a wave length of 410 millimicrons. As reference, a similar cell containing ammonium acetate solution of the same concentration as the sample solution shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the absorbance obtained shall be corrected as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, both cells shall be filled with ammonium acetate solution and the absorbance of the cell which originally contained the basic lead styphnate solution shall be measured at a wave length of 410 millimicrons. The percentage of basic lead styphnate in the sample shall be calculated on a dry basis as follows:

$$\text{Basic lead styphnate, percent} = \frac{F\,(A-B)}{W(1-.01M)D}$$

where:

A = absorbance of basic lead styphnate solution.
B = absorbance of ammonium acetate solution.
D = width of corex cell, cm.
M = moisture content of sample obtained from 4.5.3 in percent.
W = weight of sample.
F = spectrophotometric factor.. To determine the spectrophometric factor (F) weigh a 0.5 gm sample of pure lead styphnate. Transfer the lead styphnate quantitatively with 50 to 100 ml of distilled water to a 500 ml volumetric flask, add 30 ml of saturated ammonium acetate solution to the solution to the flask and dilute to mark with distilled water. Using aliquots of the solution prepare five solutions of lead styphnate ranging in concentration from .5 to 2.5 mg per 100 ml. Determine the absorbance of each solution at 410 millimicrons. Divide concentrations of lead styphnate for each of the solutions by it corresponding absorbance value. The average of the five quotients thus obtained shall be used as the spectrophometric factor (F) for the lead styphnate.

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4.5.4.2.2 "Determination of normal lead styphnate (when applicable). - Extract the residue obtained in the barium nitrate determination with sufficient 5 ml portions of ammonium acetate solution (saturated) at about 25°C. until the washings are colorless. Agitate each portion for about 1 minute, then allow each portions to remain in contact with the residue for one additional minute. Finally wash the residue with water, saturated with antimony sulfide, until the washings come through clear. Reserve the crucible and residue for the tetracene determination. Transfer the filtrate and washings quantitatively to a 250 ml calibrated volumetric flask, washing the beaker thoroughly with distilled water and dilute to mark. Using a calibrated pipette, transfer a 2 ml aliquot of this solution to a second 100 ml calibrated volumetric flask and dilute to the mark with distilled water. Transfer a portion of this solution to a corex glass spectrophotometric cell having a width of 1 cm and determine its absorbance at a wave length of 410 millimicrons. Use as reference, a similar cell containing ammonium acetate solution of the same concentration as the sample in the final dilution. Since the cell which holds the normal lead styphnate is not identical with the reference cell, correct the absorbance obtained as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, fill both cells with ammonium acetate solution, and measure at a wave length of 410 millimicrons the absorbance of the cell which originally contained the normal lead styphnate solution. Calculate the percentage of normal lead styphnate in the sample on a dry basis as follows:"

$$\text{Normal lead styphnate, percent} = \frac{F\,(A-B)}{WD}$$

where:

A = absorbance of normal lead styphnate solution.
B = absorbance of ammonium acetate solution.
D = width of corex cell, cm.
W = weight of the dried sample.
F = spectrophotometric factor. To determine the spectrophometric factor (F) weigh a 0.50 gm sample of pure lead styphnate. Transfer the lead styphnate quantitatively with 50 to 100 ml of distilled water to a 500 ml volumetric flask, add 30 ml of saturated ammonium acetate solution to the solution to the flask and dilute to mark with distilled water. Using aliquots of the solution prepare five solutions of lead styphnate ranging

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in concentration from .5 to 2.5 mg per 100 ml. Determine the absorbance of each solution at 410 millimicrons. Divide concentrations of lead styphnate for each of the solution by its corresponding absorbance value. The average of the five quotients thus obtained shall be used as the spectrophometric factor (F) for the lead styphnate.

4.5.4.3 Tetracene. - The wet residue obtained in the determination of lead styphnate shall be washed three times with alcohol, dried in an oven maintained at 60 degrees C plus or minus 5 degrees C for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125 ml beaker with a stream of water, 25 ml of water shall be added, and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the same crucible, and the residue in the crucible washed three times with boiling water, and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60 degrees C plus or minus 5 degrees C for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be caluclated as the percentage of tetracene in the sample on a dry basis as follows:

$$\text{Tetracene, percent} = \frac{(C-D)\ 100}{W(1-.01M)}$$

where:

C = weight of crucible and residue after extraction with saturated ammonium acetate solution.
D = weight of crucible and residue after treating with hot water.
W = weight of the sample.
M = moisture content of sample obtained from 4.5.3 in percent.

4.5.4.4 Antimony sulfide. - The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the sample on a dry basis as follows:

$$\text{Antimony sulfide, percent} = \frac{(D-E)}{W(1-.01M)}$$

where:

D = weight of crucible and sample after treating with hot water.

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E = weight of crucible.
W = weight of sample.
M = moisture content of sample obtained in 4.5.3. in percent.

4.5.4.5 Lead azide. - The lead azide content in the sample on a dry basis, shall be determined by subtracting from 100 the combined percentages of barium nitrate, basic or normal lead styphnate, tetracene and antimony sulfide.

4.5.5 Heat seal of sealed bag. - The adequacy of the seal on the bag shall be determined in accordance with MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

*5.1.1 Level C. - The detonators shall be packaged in accordance with dwgs 9258182 and 9258183.

*5.1.2 Level C. - The detonator parts shall be packed in accordance with MIL-STD-1169 requirements.

*5.2.1 Level C. - The detonators shall be packed as as specified on dwg 9258184.

*5.2.2 Level C. - The detonator parts shall be packed in accordance with MIL-STD-1169 requirements.

*5.3 Marking. - Detonator containers and bags shall be marked in accordance with instructions contained in dwgs 9258182, 9258183 and 9258184. Detonator parts containers shall be marked in accordance with MIL-STD-1169 requirements.

6. NOTES

6.1 Ordering data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article samples.

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c. Provisions of MIL-A-48078.

6.2 Submission of inspection equipment designs for approval. - (See MIL-A-48078). Submit equipment designs as required, to Commander: Picatinny Arsenal, ATTN: SARPA-QA-T Dover, New Jersey 07801.

6.3 Distribution of ammunition data cards. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M Dover, New Jersey 07801.

6.4 Check test for possible deterioration. - The following test is designed to prevent detonators that have deteriorated from being assembled into major ammunition items. The test shall be performed on detonators from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are assembled does not require that the check test be performed, the Contracting Officer should insert in the contract a provision for the performance of the check test. The Contracting Officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. Detonator assemblies shall be selected from each lot for this test.

6.4.1 Functioning. - A sample of 315 detonators shall be selected at random from each lot. If 4 or more detonators fail to comply with the requirement specified, the lot shall be rejected. If more than one defective is found, a second sample of 315 shall be selected at random and tested. If in the combined first and second sample a total of 5 or more defectives are found the lot shall be rejected. The test shall be performed as specified in 4.5.2.1 using test equipment as specified 4.4.4.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccurances in these notations. Bidders and contractors are cautioned to evaluate content irrespective of the marginal notations and relationship to the last previous issue.

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6.6 Supersession Data. - This specification includes the requirements of MIL-D-46210A (MU) dated 25 January 1965.

Custodian:
Army-PA

Preparing Activity:
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